



CP5046 Individual Video Defence

Title: Comparative Analysis & Game Prototyping Defence

Team Name: Group 1

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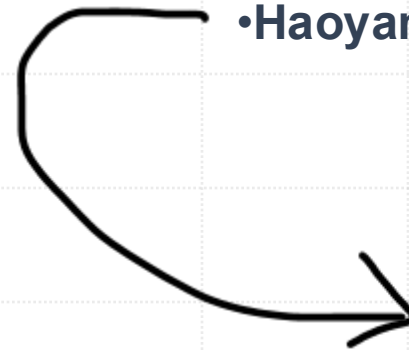
Project Scope & Team Roles

Target & Deliverables

- Target Audience: Kids aged 4–6 years old.
- Alpha Phase: 12 interactive game levels (3 levels per member).
- Beta Phase: 1 core concept with 9 custom garden themes.

Team Division

- Weichao Sun: Comparison game and team leader
- Xianzhe Chen: Animal Matching Game
- Bohong Wang: Shape & Number Games
- Haoyang Liu: Sorting Games



Is me!

Development Approach Comparison

Waterfall Limitations

- Rigid Assumptions
- Unsuitable for Kids
- Late Risk

Agile Scrum Advantage

- Iterative Cycles
- Prioritized Backlog
- Early Playtesting



Why Projects Fail & Our Mitigations

Process Pitfalls

- "Mechanical Scrum"
- Outputs vs. Outcomes

Smart Mitigations

- Active Backlog
- Empirical Testing
- Strict DoD



Design Thinking

1. Empathise

2. Prototype

3. Adaptability



Sprint Pilot 1: Fruit Sorting

- **Design & Trade-offs**
- **Goal:** Sort apples, bananas & grapes into matching color bins.
- **UX Feature:** Bouncing start button eliminates text reading.
- **Trade-off:** Used static background art to save sprint time.



Sprint Pilot 2: Alphabet Matching

- **Phonics Scaffolding**
- Goal: Pair cards (Apple, Bee, Car, Dog) to letters (A, B, C, D).
- Reward UX: Instant score popups motivate players.
- Team Fix: Expanded drop zone size after Daily Scrum feedback



Sprint Pilot 3: Color Laboratory

- **Cognitive Challenge**
- **Goal:** Sort mixed items (animals, cars, plants) purely by color.
- **Distractors:** Yellow chicks and red cars test focus skill.
- **Data Metric:** Avg. 2.3 attempts showed strong retention.





Scrum Team Reflection

Core Pillars

Transparency, Inspection & Adaptation.

Strict time-boxing ensured MVP delivery.

DoD guaranteed bug-free interaction.

Key Insights

Value learning outcomes over raw lines of code.

Retrospectives improved internal teamwork.

Standups focused strictly on operational blockers.



Conclusion

Key Achievements

Delivered 12 Alpha educational game levels.

Built 3 interactive sorting modules successfully.

Proved Agile + Design Thinking reduces risk

Continuous Improvements

Apply "5 Whys" in retrospectives.

Run early technical spikes for asset scaling.

Maintain continuous portfolio reflections.



Thank you for listening